

Close

**JGME-D-25-01151**

## **"Algorithmic Scheduling to Improve Resident Well-Being and Equity in Call Distribution"**

**Original Submission**

**Matthew A. Porter, B.S. (Reviewer 1)**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Reviewer Recommendation Term:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Major Revision        |
| <b>Overall Reviewer Manuscript Rating:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 60                    |
| <b>Custom Review Question(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Response</b>       |
| Please rate the manuscript based on if the topic may be of interest to program directors or GME educators, researchers, or deans.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4=moderately agree    |
| Please rate the manuscript based on if the conclusions are supported by data or evidence presented.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2=moderately disagree |
| Please rate the manuscript based on if the manuscript needs additional statistical review.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3=agree               |
| <b>Comments to Editor:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                       |
| <p>*Please Click on "Reviewer Instructions" at the top of this screen and read before answering the following.</p> <p>Is there a financial or other conflict of interest between your work and that of the authors?</p> <p>YES ___ NO _X_ If yes, please contact the Editor-in-Chief.</p> <p>Confidential comments to the Editor: Please make comments here to the Editor, which will not be passed on to the authors:</p> <p>Confidential Comment 1</p> <p>Thank you for the privilege of reviewing this manuscript. I am slightly concerned that the authors frame their work as filling a gap in GME, stating "Algorithmic scheduling, while common in operations research, has rarely been applied within graduate medical education" (Page 1, lines 36-38). However, the manuscript entirely omits discussion of existing commercial resident scheduling platforms already deployed in residency programs, including Amion, QGenda, Lightning Bolt, and Aurora-Med—several of which incorporate optimization algorithms, fairness metrics, and ACGME compliance features. The authors provide no justification for building custom Python/OR-Tools software rather than evaluating or adopting existing commercial solutions, nor do they discuss potential advantages of their approach versus established platforms. While this does not diminish the technical merit of their work, programs reading this manuscript would benefit from understanding the rationale for custom development versus commercial adoption, including considerations of cost, maintenance burden, and comparative functionality. Brief acknowledgment of the existing landscape would strengthen the contribution and help readers contextualize this work within the broader GME scheduling ecosystem.</p> |                       |
| <b>Comments to Author:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                       |
| <p>Comments for authors: Present general comments first, followed by specific points. Include page and paragraph citations, if applicable, to assist authors. Please NUMBER each of your comments, as the authors will be asked to respond to each numbered comment.</p> <p>General Comments:</p> <p>This manuscript addresses an important operational challenge in graduate medical education. The authors' decision to implement their solution using open-source optimization software (Python-based Google OR-Tools) rather than proprietary AI models or commercial platforms deserves recognition and emulation. This approach offers several advantages: zero software licensing costs, full transparency of the optimization logic, and complete reproducibility for other programs with basic programming resources. The constraint-based optimization method provides explainable, deterministic results without the privacy and cost concerns associated with conventional software and a growing number of AI "solutions". However, the manuscript would be strengthened by explicitly acknowledging this design choice and discussing the trade-offs between novel</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                       |

implementation of open-source solutions versus commercial scheduling platforms already available to residency programs (see specific comments below). With revisions addressing reproducibility details and proper contextualization within the existing GME scheduling landscape, this work could provide valuable guidance for programs considering cost-effective, transparent scheduling solutions.

## Specific Comments

### Comment 1

\* Page 1 Lines 39-45 & Lines 52-59

The manuscript claims the authors "developed and implemented a constraint-based scheduling software" (Page 1, lines 40-41), but the Methods clarify they implemented Google OR-Tools CP-SAT (Page 1, lines 52-53)—an established, publicly available optimization solver. This language misrepresents the authors' contribution. The actual work involved applying existing optimization software to psychiatry residency scheduling: defining domain-specific constraints, formulating the problem for CP-SAT, implementing the interface, and validating the approach. These contributions are meaningful but should be distinguished from software development. Fine-grained specificity on programming language is essential. The OR-Tools documentation from the manuscript's cited first reference (<https://developers.google.com/optimization>) explicitly highlights "C++, Python, C#, or Java," while the OR-Tools GitHub repository (<https://github.com/google/or-tools>)—also linked on that same front page—shows the codebase includes C++ (79.1%), Python (6.3%), Julia (3.2%), Starlark (2.2%), C# (2.1%), and Java (2.0%), with multiple distribution channels including source builds, PyPI for Python, NuGet for C#, and Maven Central for Java. This diversity of implementation options could confuse teams seeking to replicate the authors' methods. The Methods must specify: whether the implementation used pre-compiled packages (e.g., PyPI: `pip install ortools`) or built from source, the specific Python version (recommend Python 3.9+ per Google maintainers for current OR-Tools compatibility, or exact version like "Python 3.10.x"), Google OR-Tools version (e.g., "OR-Tools v9.12.xxxx"), computational environment (OS, hardware specifications), and additional dependencies. Without these specifications—particularly the installation method and exact version details—other programs cannot reliably replicate the approach or may encounter compatibility issues.

### Comment 2

\* Page 2 Lines 20-23

The manuscript introduces "Gini and Temporal Gini stratified by PGY level" without justifying why these metrics were selected or how they relate to call scheduling fairness. Add rationale: "The Gini coefficient quantifies inequality in call distribution, where values near 0 indicate equitable distribution. Temporal Gini assesses spacing between shifts, with lower values indicating uniform temporal distribution and reduced clustering." Cite prior use in scheduling or workforce literature, or explicitly state if this represents novel application to residency scheduling.

### Comment 3

Page 2 Lines 1-10 and Table 1

Table 1 describes constraints qualitatively but lacks quantitative parameters required for reproducibility. The "Soft optimization goals" section mentions minimizing inequality and maximizing spacing without specifying weighted penalties or mathematical formulation. CP-SAT performance depends directly on constraint weighting and formulation. Add to Table 1 or supplementary material: weights for each soft optimization goal (e.g., "Gini weight = X, temporal spacing weight = Y"), mathematical formulation or penalty functions, thresholds or target values (e.g., minimum days between calls), and relative prioritization when constraints conflict. For hard-coded rules, specify numerical values (e.g., "minimum 10-hour rest period post-call" not just "rest protection after call").

### Comment 4

Page 3 Lines 25-27

The claim of "25-35% relative improvement in equity" (lines 26-27) lacks explanation of the calculation method. Explicitly demonstrate: does "relative improvement" mean percent reduction from baseline ( $\Delta\text{Gini}/\text{Baseline Gini} \times 100\%$ ), movement toward perfect equity ( $\text{Gini} = 0$ ), or another interpretation? For example, if  $0.073/0.110 \times 100\% = 66\%$  reduction  $\approx 35\%$  relative improvement, state this clearly. Without transparent methods, readers cannot verify claims or apply metrics to their data.

### Comment 5

Page 3 Lines 31-34

The manuscript acknowledges "small sample size skewing formal statistical testing" (lines 32-34) but claims "consistent and significant improvement" despite PGY-4 Gini CI crossing zero ( $-0.038 \rightarrow 0.000$ ) as shown in Table 2. This contradicts statistical significance criteria. Clarify "small sample size" (N residents per PGY, number of calls, or both?). Conduct post-hoc power analysis. Replace vague language: state "The small sample size (N = [specify]) limited statistical power, particularly for PGY-4 where [specify N and calls]." Acknowledge PGY-4 Gini improvement was not statistically significant rather than grouping it with "consistent improvement."

### Comment 6

Page 4 Lines 39-50 and Throughout

The manuscript alternates between "software," "scheduling program," "algorithm," "algorithmic approach," and "constraint-based scheduling software" without standardization. Statements like "the new software reduced schedule-building time from approximately 30 hours per block (manual) to less than 3 hours" (lines 39-40) and

"the algorithm automatically applied all eligibility and fairness constraints" (lines 41-42) lack technical detail. Standardize terminology. Choose one term ("scheduling solution," "optimization algorithm," "scheduling tool") or clarify this is Python script-based implementation. Specify implementation format: standalone Python script via command line, Jupyter notebook, third-party GUI (e.g., VSCode/Neovim/Zed/Cursor/PyCharm), web application (ChatGPT/Grok/Gemini), or if everything was done with Microsoft Excel. Describe workflow: how inputs are provided (CSV file, spreadsheet), how OR-Tools CP-SAT is invoked, output format and accompanying metrics, and actual runtime (you mention "minutes" on Page 2, line 13). Clarify how constraints are encoded (Boolean expressions in CP-SAT model), whether constraints are hard-coded or configurable via input files, and how program-specific rules are specified and modified.

Comment 7

Page 5 Lines 42-58

Q2 showed a 6-percentage-point decrease in personal needs accommodation (75% → 69%)—the only negative result—yet is framed as "Interestingly" (line 42) rather than acknowledged as a limitation. The explanation that manual schedulers "would likely have various informal personal pieces of information about their co-residents" (lines 50-53) is speculation without data on what informal information was used or whether residents preferred informal accommodation. No discussion addresses whether chief residents were comfortable making informal accommodations or concerns about fairness. No mitigation strategies proposed (manual overrides, expanded preference capture, hybrid approaches). The text states "the new software does not have access to outside of formally requested PTO and non-call requests" (lines 54-56) but offers no solution. Please consider reframing: "Satisfaction with personal needs accommodation decreased 6 percentage points (75% to 69%), the only decline among surveyed domains. While the algorithmic approach ensures transparent, equitable treatment via formal PTO and non-call requests, manual schedulers may have incorporated residents' personal circumstances not captured through formal channels, raising questions about trade-offs between objective fairness and individualized flexibility. Future iterations should: (1) expand formal preference mechanisms, (2) implement hybrid approaches allowing limited manual adjustments, or (3) survey residents about specific personal needs better addressed under manual scheduling."

Comment 8

Page 6 Lines 19-26

The Conclusion introduces new terminology ("constraint-based algorithmic scheduling program" at line 19), uses vague language ("more optimized" at line 19), and makes unmeasured claims. "Protecting didactic time" (line 26) was listed in Table 1 as a soft optimization goal but never evaluated in Results—no data show whether didactic requests were respected or improved. "Systems-based practice goals" (line 26) is never defined or measured. "Aligns with resident well-being" (line 25) overstates findings given Q2 decreased and Q6 improved minimally (4 percentage points). "Feasible for adoption by other programs" (line 24) provides no evidence on required programming expertise, costs (licenses, development, maintenance), transferability to different specialties or program sizes, implementation barriers, or technical infrastructure. This single-site QI project in one psychiatry program during 2025-2026 presents findings as broadly generalizable. "Substantially less administrative effort" (line 21) counts only 3 hours to run the scheduler, ignoring upfront development investment and ongoing maintenance. Revise to acknowledge limitations and provide realistic adoption considerations, including: programming expertise requirements, development/customization time, ongoing maintenance, trade-offs between algorithmic fairness and individualized needs. Future multi-site studies should evaluate generalizability across specialties, program sizes, and call structures over multiple years. Explicitly measure stated goals like didactic time protection.

| Transfer Authorization                                                                                                                                         | Response |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| If this submission is transferred to another publication, do we have your consent to include your identifying information?                                     | Yes      |
| If this submission is transferred to another publication, do we have your consent to include your review?                                                      | Yes      |
| If this submission is transferred to another publication with "Open Peer Review", do we have your consent to publish your review in a pre-publication history? | Yes      |

Close